

— PUTNAM PROBLEM

Determine all positive integers n for which there exist $n \times n$ matrices A_1, A_2, A_3, A_4, A_5 with real entries such that

$$\begin{aligned} A_i^2 &= 0 && \text{for every } i, \\ A_i A_j + A_j A_i &= I && \text{for all } i \neq j. \end{aligned}$$

Here I is the $n \times n$ identity matrix.